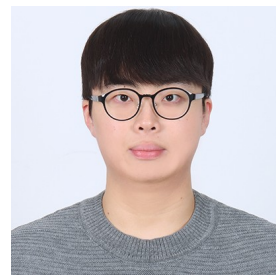


YOUNGCHAN LEE | RESUME

- » Status: Undergraduate Student at Sungkyunkwan University
- » Skills: Python, YOLOv11, OpenCV, Excel, PowerPoint
- » Interests: Smart Factory, Video AI, Manufacturing Process Optimization
- » Activities: Fitness, Basketball, Football, Music



» » » Summary

Undergraduate student in Chemical Engineering and Polymer Science at Sungkyunkwan University, with experience in AI-based smart factory research, manufacturing process analysis, and video-based worker behavior recognition. Participated in a capstone design project on automated Line-of-Balance (LOB) analysis for manufacturing packaging lines using YOLOv11-based object detection, SORT/DeepSORT tracking, GRU-based time-series action recognition, and real-time dashboard visualization.

» » » Experience

- | | | |
|--|--------------------|--|
| '26/03 - '26/06 | Participant | Smart Factory Capstone
Design project |
| <ul style="list-style-type: none">» Conducted a capstone design project titled Automated Line-of-Balance (LOB) Analysis for Manufacturing Packaging Lines Using YOLO-Based Video AI» Developed a YOLOv11-based object detection system to detect Person, Hand, Hat, and Product in manufacturing packaging-line videos» Applied SORT and DeepSORT tracking algorithms to maintain worker-specific Track IDs across video frames» Built a GRU-based time-series action recognition model to classify worker actions into Pick, Place, Working, and Other categories» Calculated Handling Time and Cycle Time using RoI entry/exit rules, $\pm t$ time-window matching, and Track-level debouncing» Automated LOB analysis by applying worker action and process-time data to LOB formulas and visualized real-time results through a dashboard» Produced research outcomes including a patent and software registration titled Real-Time LOB Calculation Program Using YOLO- and GRU-Based Time-Series Action Recognition» Prepared a research paper titled A Study on Worker Detection and Automated Line-of-Balance (LOB) Analysis in Manufacturing Packaging Lines Using YOLO-Based Video AI» Conducted a team-based literature review and presentation video on prior studies including YOLO-DS, worker anomaly detection using human-object interaction, SMA-YOLO, YOLOv11, PCB defect detection with anomalous trend alarming, LIVOS, and adaptive human-centric video anomaly detection | | |

» » » Education

- | | | |
|--|---|----------------------------|
| 2021/03 - now | B.S. in Chemical Engineering and Polymer Science | Sungkyunkwan
University |
| <ul style="list-style-type: none">» Completed core coursework in Organic Chemistry, Physical Chemistry, Thermodynamics, Separation Processes, and Heat and Mass Transfer» Studied fundamental chemical engineering principles related to reaction systems, phase behavior, transport phenomena, and process analysis» Developed a strong academic foundation for understanding industrial manufacturing processes and process optimization» Applied chemical engineering knowledge to analyze process flow, operation time, and productivity improvement in smart factory-based manufacturing systems | | |